

EE4437 Assignment 2026

Please read these instructions carefully

1. **Submission:** Please submit your assignment to [Canvas > Assignments > Homework Assignment for Part II](#). Hard copy submission, e-mail submission, or submission to the wrong assignment/folder will not be considered.
2. **Deadline:** [Sunday, 19 April 2026 \(11 pm\)](#). Late submission will incur a penalty. No marks will be awarded for submissions after the solutions have been released.
3. **Format:** Submit as a single pdf file. If you are using e.g. Microsoft Word, or scanning your solutions as images, convert and combine them into a *single PDF file* before submission.
4. **File Name** format is: [\[Matric no.\]_\[Name\].pdf](#), e.g.: [A012345B_John_Tan.pdf](#).
5. **Resubmission:** If you wish to make changes after submission but before the deadline, delete the old submission and submit the modified version. If a student has multiple submissions under his/her name, [only the latest version submitted before the deadline will be considered](#).
6. **No compressed files:** Zipped or compressed files will [not be considered](#).
7. **Answer all questions.** The total mark is [30](#). This assignment carries a CA weightage of [10%](#).
8. **Please show your complete workings.** Marks may be deducted without complete workings even if the final answer is correct.
9. You may make use of the following:

Electron charge, $e = 1.6 \times 10^{-19}$ C

Permittivity of vacuum, $\epsilon_0 = 8.85 \times 10^{-12}$ F/m

Boltzmann constant $k_B = 1.38 \times 10^{-23}$ J/K

Thermal energy, $k_B T = 25.9$ meV at temperature of 300 K

Mass of electron in vacuum, $m_e = 9.1 \times 10^{-31}$ kg

Planck's constant, $h = 6.63 \times 10^{-34}$ J s

Reduced Planck's constant, $\hbar = 1.05 \times 10^{-34}$ J s

$E(\text{eV}) = 1.24/\lambda(\mu\text{m})$

Question 1

- (a) A boron-doped Si photoconductor has a resistance of $160\ \Omega$ in the absence of infra-red (IR) radiation. When IR radiation of $20\ \text{mW/m}^2$ is incident on the photoconductor, the resulting photocurrent constitutes 75% of the total current at steady state. Calculate the resistance of the photoconductor at steady state. (5 marks)
- (b) The photoconductor in Q.1(a) is then exposed to infra-red (IR) radiation of $50\ \text{mW/m}^2$. Calculate its new resistance at steady state. (5 marks)
- (c) Optical radiation of $70\ \mu\text{W}$ power is incident on an open-circuited pn photodiode. The photodiode attains an open circuit voltage responsivity of $3.6 \times 10^3\ \text{V/W}$ at steady state. If the negative saturation current of the photodiode is $1.5\ \text{nA}$ and the wavelength of the incident light is $650\ \text{nm}$, calculate the quantum efficiency of the photodiode. You may assume room temperature of $300\ \text{K}$. (6 marks)

Question 2

A quantum well infra-red photodetector consists of alternate layers of semiconductors A and B, the dimensions and material parameters of which are given in the table below.

	A	B
Thickness of each layer	12 nm	20 nm
Band gap energy	0.73 eV	1.86 eV
Effective electron mass	$0.08\ m_e$	$0.28\ m_e$

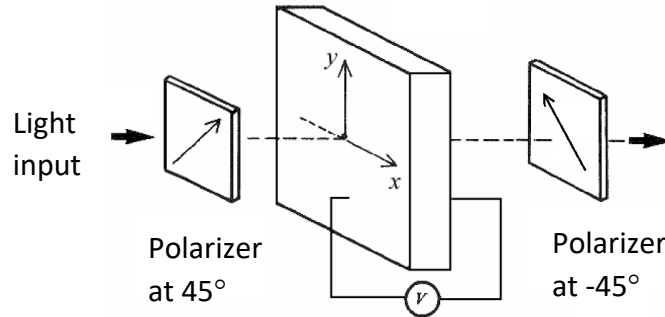
(N.b.: m_e is the electron mass in vacuum).

If the quantum wells formed in the valence and conduction bands are of equal depth, evaluate the following:

- (a) Estimate the wavelength corresponding to the main absorption peak of the QWIP using the infinite quantum well model. State any assumptions in your calculations (other than the ones stated in the question). (6 marks)
- (b) Is your estimated wavelength longer or shorter than the actual wavelength? Explain your answer. (3 marks)

Question 3

The figure below depicts an electro-optic crystal modulator placed between two parallel polarizers, oriented at $\pm 45^\circ$ to the x and y principal axes of the modulator. The refractive index n_0 of the crystal in the absence of electric field is 2.7, while its Pockels coefficient is 2.4×10^{-11} m/V along the x -axis and -2.4×10^{-11} m/V along the y -axis. The light at the input is originally unpolarized and has a wavelength of $\lambda_0 = 550$ nm in vacuum.



Determine the phase difference between the components of light polarized in the x and y directions after passing through the crystal. Express your answer in terms of V , the voltage applied across the modulator.

(5 marks)

End of Assignment